

Exit Through the Candy Shop (2022)

In collaboration with the online investigative collective Bellingcat, I am currently developing network analysis tooling for researching suspicious companies and directors through the UK Companies House register. I am undertaking this project within a funded tech fellowship provided by Bellingcat, supported by the Alfred Landecker Foundation.

My interest in navigating corporate data was heightened following reporting on a collection of American candy shops that sprung up in recent years on post-pandemic Oxford Street in London (Figure 1). Once the UK's most prominent retail destination, the street is now home to a number of candy shops that have attracted media attention for tax evasion and the sale of counterfeit goods, from unsafe vapes to fake Rolexes and Wonka chocolate bars. Network analysis of directors and addresses reveal a complex web of shell companies registered in locations often used by other suspicious directors or known criminals (Figure 2).

I am developing this tooling to address existing needs within the open-source intelligence (OSINT) research community by improving the ease with which large networks of companies can be navigated.

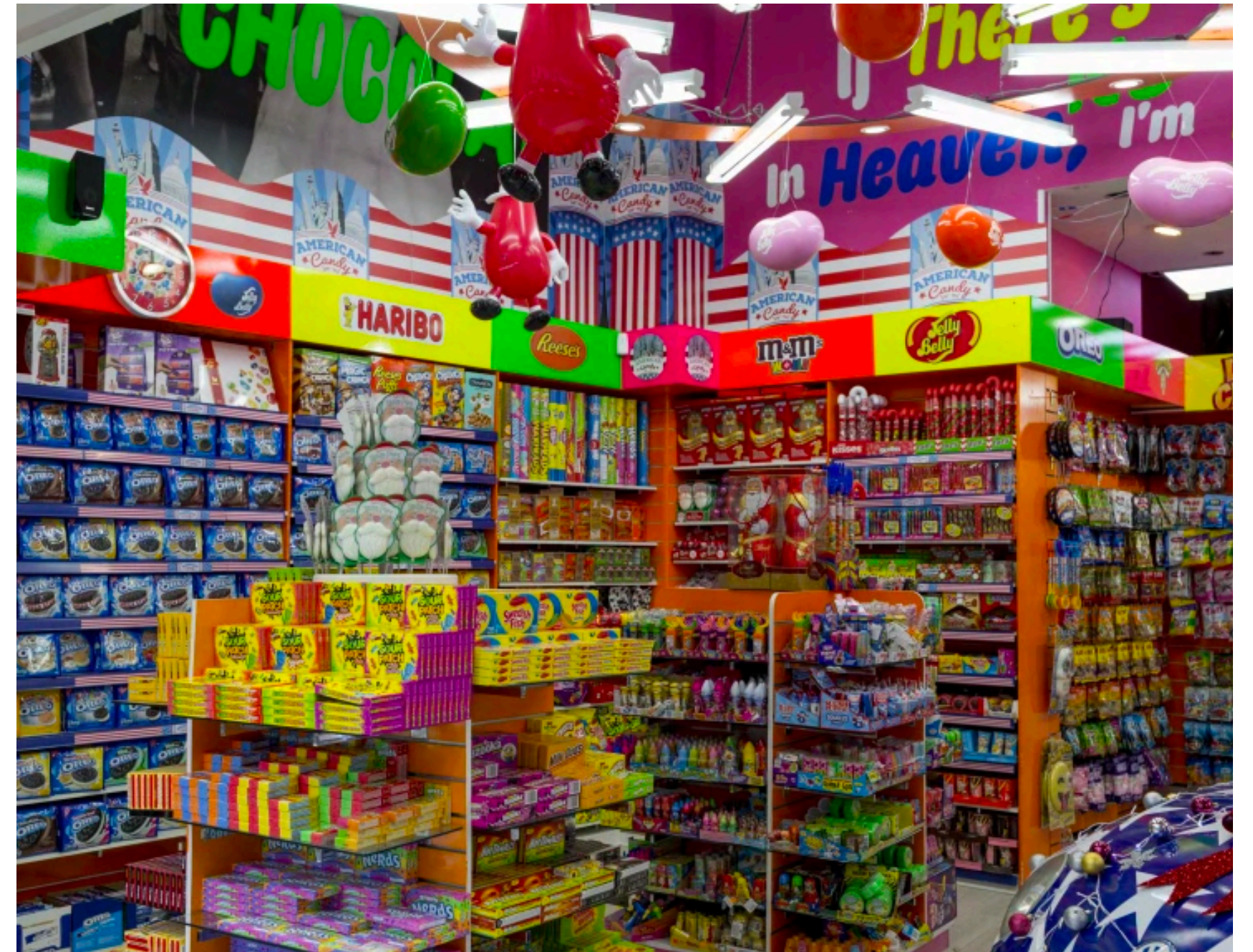


Figure 1: American candy store from Richard Baker / In Pictures via Getty Images



Figure 2: 'Hold Everything' is a virtual office to over 4000 businesses at 207 Regent Street

An initial prototype of the software enables users to explore company addresses connected by common directors. For example, given the address for 'Kingdom of Sweets' on Oxford Street as input, the tool returns a number of locations with thousands of businesses registered to each one, within a mere 2 degrees of separation from 'Kingdom of Sweets' (Figure 3). These extraordinarily popular addresses are often virtual offices or mail-forwarding services for shell companies, which have proven popular with fraudsters. Applying a news search filter across the tens of thousands of addresses, companies and directors reveals a selection of well known instances of financial crime:

- Viceroy Jones New Tech Ltd. registered at 20-22 Wenlock Road: £9m truffle farm investment scam targeting pensioners.
- UK Services and Support Limited registered at 207 Regent Street: £1.6 million passport scam.
- Westland Securities registered at 207 Regent Street: Sibling duo who hacked Italian prime minister and other elites.
- Pompolo Limited registered at 2 Woodberry Grove: Paul Manafort (Donald Trump's ex-campaign chair) found guilty of tax evasion and embezzlement via offshore companies.
- Blythe (Europe) Ltd registered at No. 29 Harley Street: linked to assets of Viktor Yanukovich, former corrupt president of Ukraine.
- Helve TCS Limited once registered at 207 Regent Street: Julian Beadsley was part of a gang convicted of defrauding £2 million from over 100 victims through a boiler room investment fraud scheme.
- Hillgold Properties once registered to 2 Woodberry Grove: Russian crime syndicate including jailed Alexei Anokhin (nephew of Russia's ex-deputy minister of defence Tatiana Shevtsova) defrauded HMRC of 1 billion rubles equivalent.
- RISC Management once registered at 788-790 Finchley road: private investigator firm accused of bribing the Metropolitan Police Service. The firm was created by Stephen Curtis, a lawyer connected with Russian oligarchy who died under suspicious circumstances in 2004.
- Elite Energy Saving Limited registered at 207 Regent Street: director Stuart Ralph Poppleton is a known alias of notorious disqualified print farmer Neill Stuart Malcolm John.
- Valemont Properties Limited registered at 63-66 Hatton Garden: received £142m through 'sham litigation' in Moldovan courts as part of a scheme labelled by the Organised Crime and Corruption Reporting Project as "the Global Laundromat".
- Globex International LLP once registered at 788-790 Finchley Road: Azerbaijani gold fields worth billions of dollars were transferred to President Ilham Aliyev's family members through this company.

[Code](#)

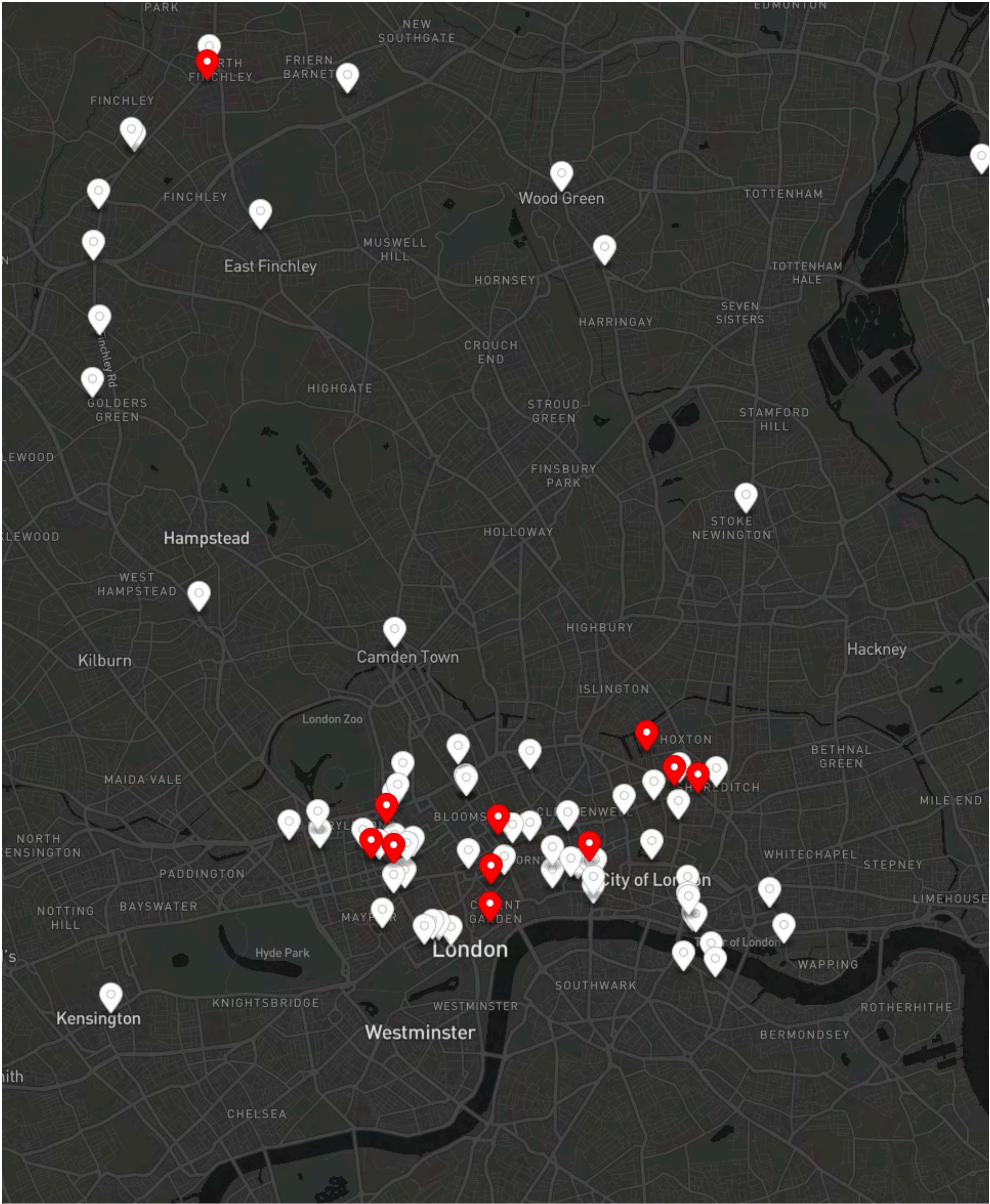


Figure 3: Locations with over 1000 businesses registered to each. Every location is within 2 degrees of separation from the 'Kingdom of Sweets' address. Red addresses have registered businesses with known fraudulent activity.

Curation Cartography (2022)

For several years internet radio has taken over as my primary means for discovering new music. I find the format to be more surprising and taste-shifting in comparison to alternatives like Spotify that rely heavily on recommendation algorithms. The experience of transitioning from a state of disorientation to recognition of an artist or DJ’s style over several episodes is addictive. I have been prototyping ways in which this metadata rich format could further differentiate as a uniquely valuable engine for music discovery.

I have begun developing software to visualise how different shows from NTS Radio are connected through record labels (Figure 4). Record labels are powerful curators yet often remain obscured from the listener. Processing hundreds of hours of mixes to visualise the curatorial pathways, opens up new possibilities for discovering artists, collectives and scenes that might otherwise remain submerged in the endless stream of sound.

I’m currently in friendly dialogue with NTS Radio’s tech team exploring how the tool could be further developed for artists, users and internal use-cases.

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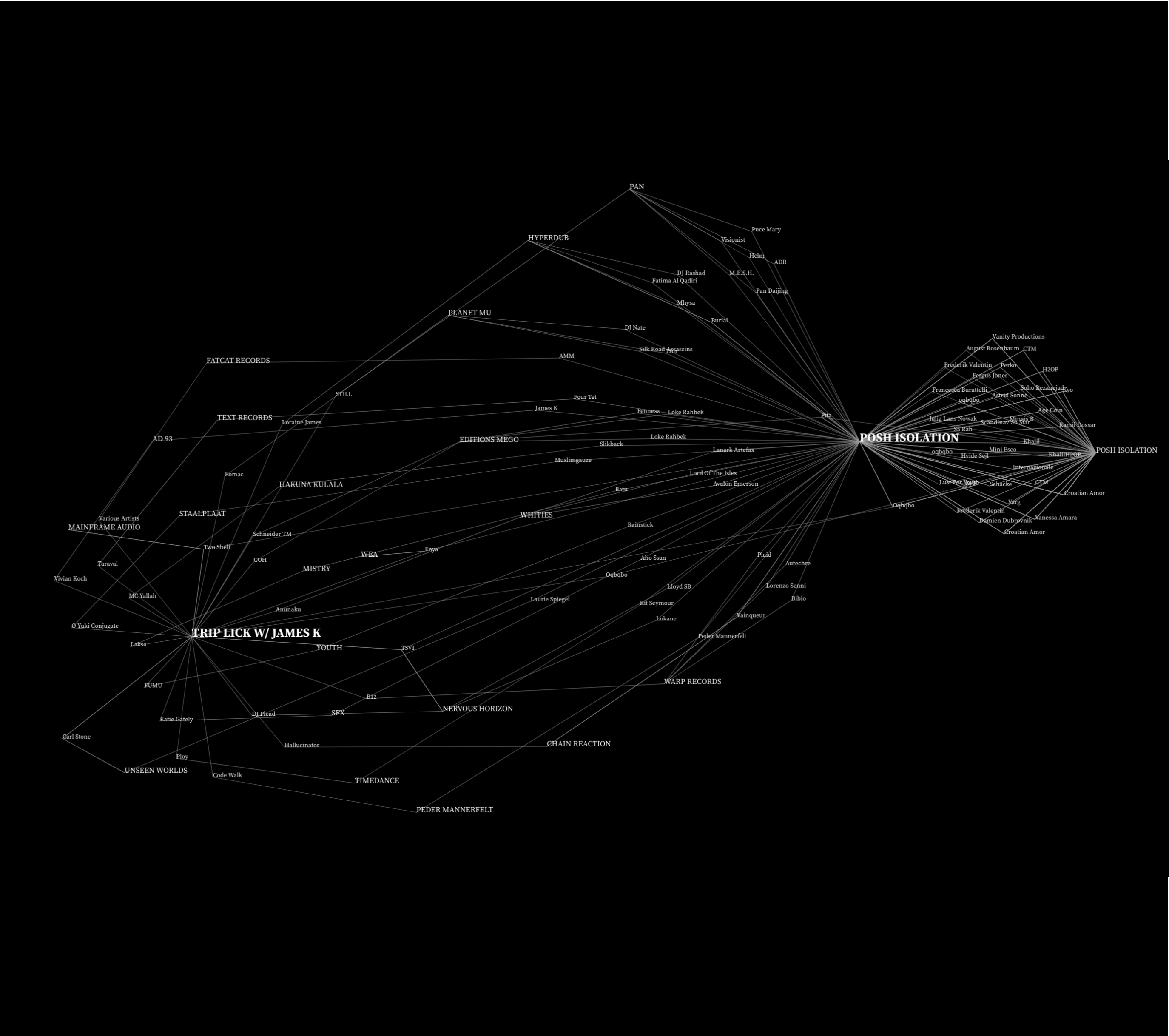


Figure 4: Visualisation of artist/label connections between two shows, ‘Trip Lick w/ James K’ (14 episodes) and ‘Posh Isolation’ (67 episodes), cultivated over 80+ hours of radio transmission

Inpainting the Archive (2022)

Inpainting is the task of filling in the missing regions of an image with content. Denoising Diffusion Probabilistic Models (DDPM) represent an emerging paradigm in generative modelling and have been shown to outperform state of the art GAN-based approaches to inpainting (Lugmayr et al., 2022). I was curious to explore how this technique could be repurposed within a playful research tool for combining fragments of conceptually linked imagery and generating images that prompt novel associations. How can fragments of images act like spores in culture, growing and composing new forms in semantic and aesthetic harmony, providing the viewer with new environments to explore?

Inspired by the anchors black-box model interpretability algorithm (Ribeiro et al., 2018) for visualising the contribution of regions within an image towards a machine learning model's classification, I defined an alternative system leveraging Wikimedia's extensive image archive that combines regions of images to generate any number of ambiguous and unclassifiable images. I think it's interesting how pieces of images can be used to reinforce binary classifications whilst also acting as portals to an infinite array of new images and potential associations. The system works as follows:

1. User inputs the title of a Wikipedia page and a custom image search function returns a selection of images taken from backlinks, thus providing a diverse dataset of imagery associated with the input in often unusual and surprising ways (Figure 5).
2. User selects images of interest from the dataset which are each segmented into jigsaw like regions using a segmentation algorithm.
3. User composes a partial image from these regions.
4. A mask is generated from the partial image.
5. Masked regions in the image are filled with content using a DDPM trained on ImageNet implemented in PyTorch (Lugmayr et al., 2022) (Figure 6).
6. System outputs the generated image and a nano detailing the user generated composition of image regions and the Wikipedia pages they were taken from (see next page).

The images generated by this systems are messy, confused and hallucinatory, revealing biases in the training data as the model processes unusual compositions of fragments; dogs continue to bubble up in the soupy landscapes. However these 'fails' or 'mistakes' continue to offer interesting occasionally surprising insights. With more time I look forward to integrating much higher fidelity AI inpainting software within this system such as Stable Diffusion.

Inspiration for this system was developed over a number of evenings leading a Creative Coding meet-up during a programming retreat at the Recurse Center in NYC.

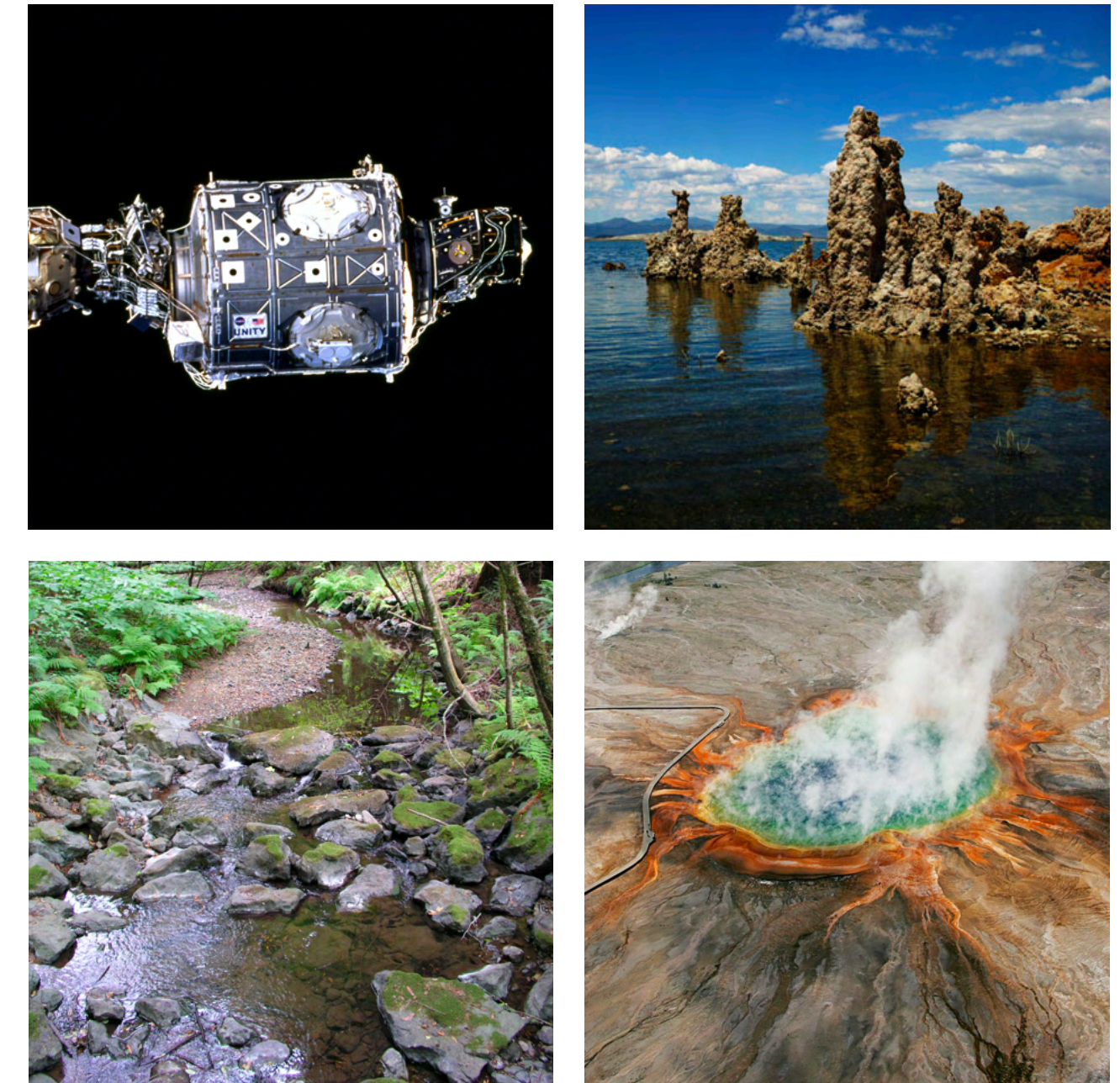


Figure 5: Selection of images from Wikipedia pages linking to the Wikipedia entry for 'Extremophile'

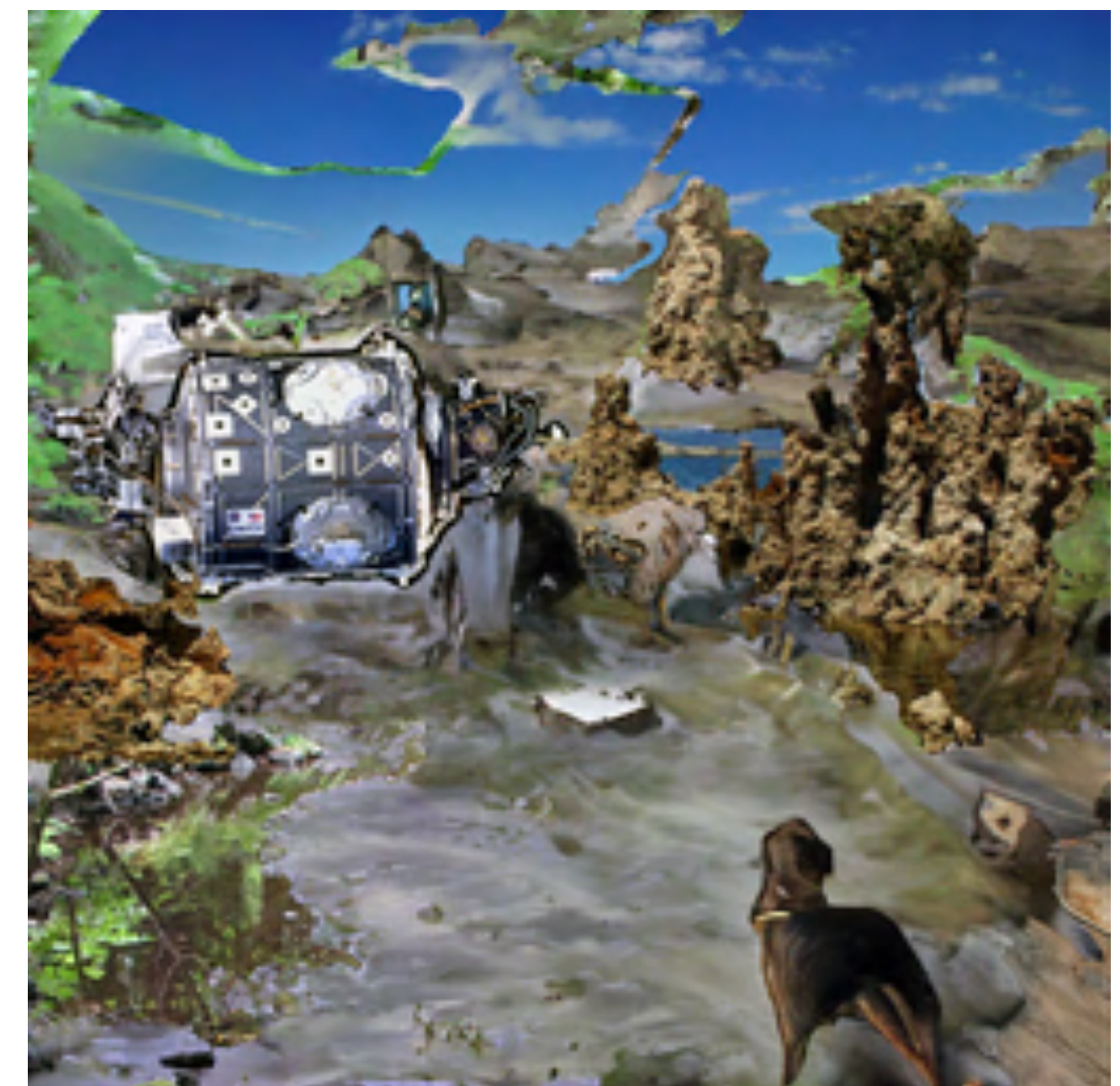
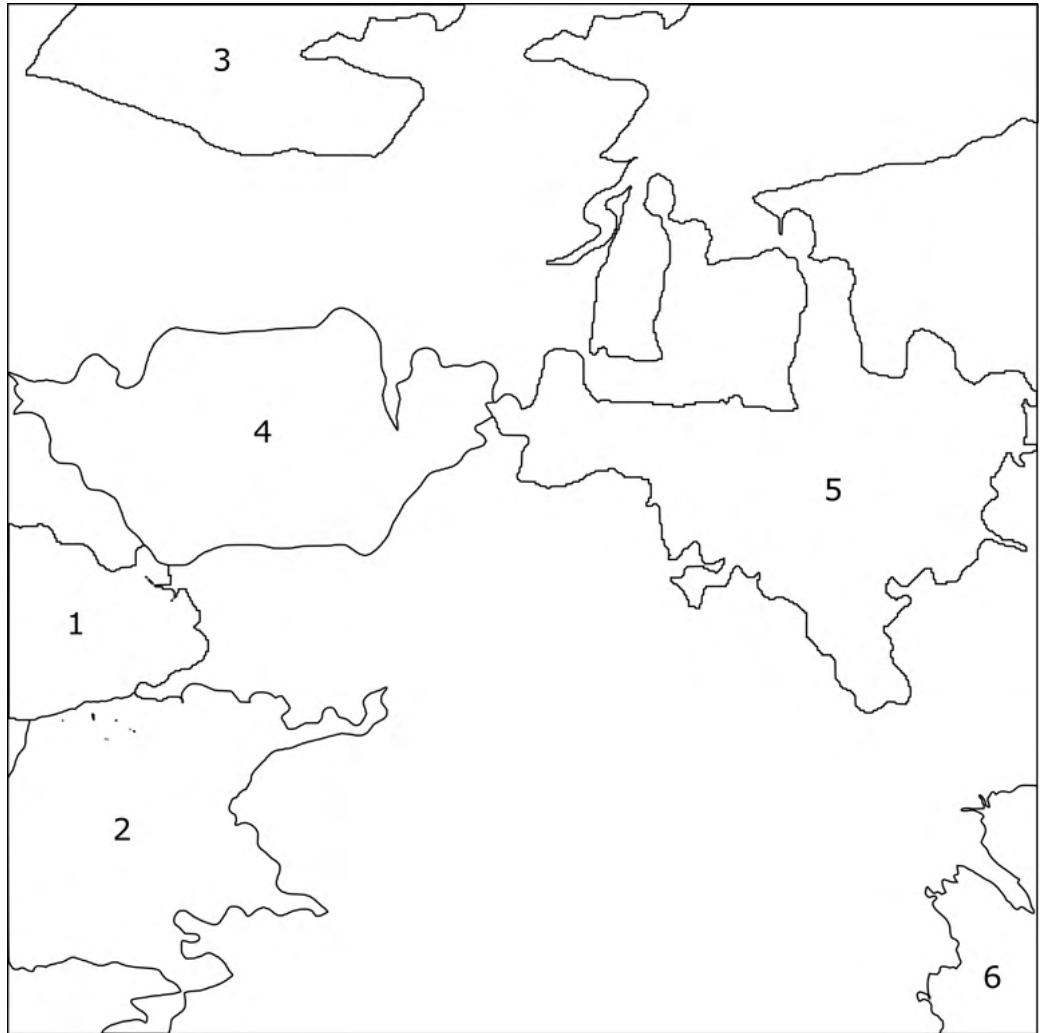
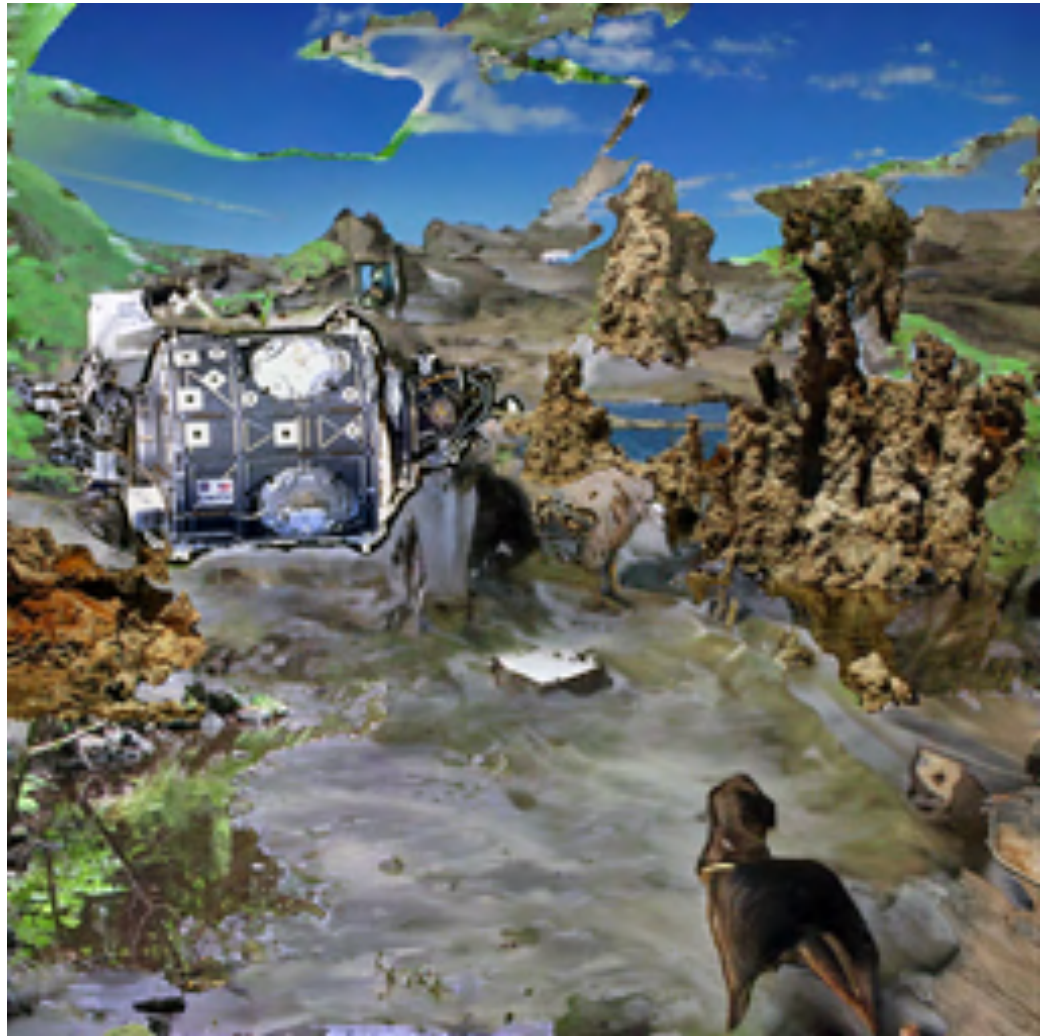
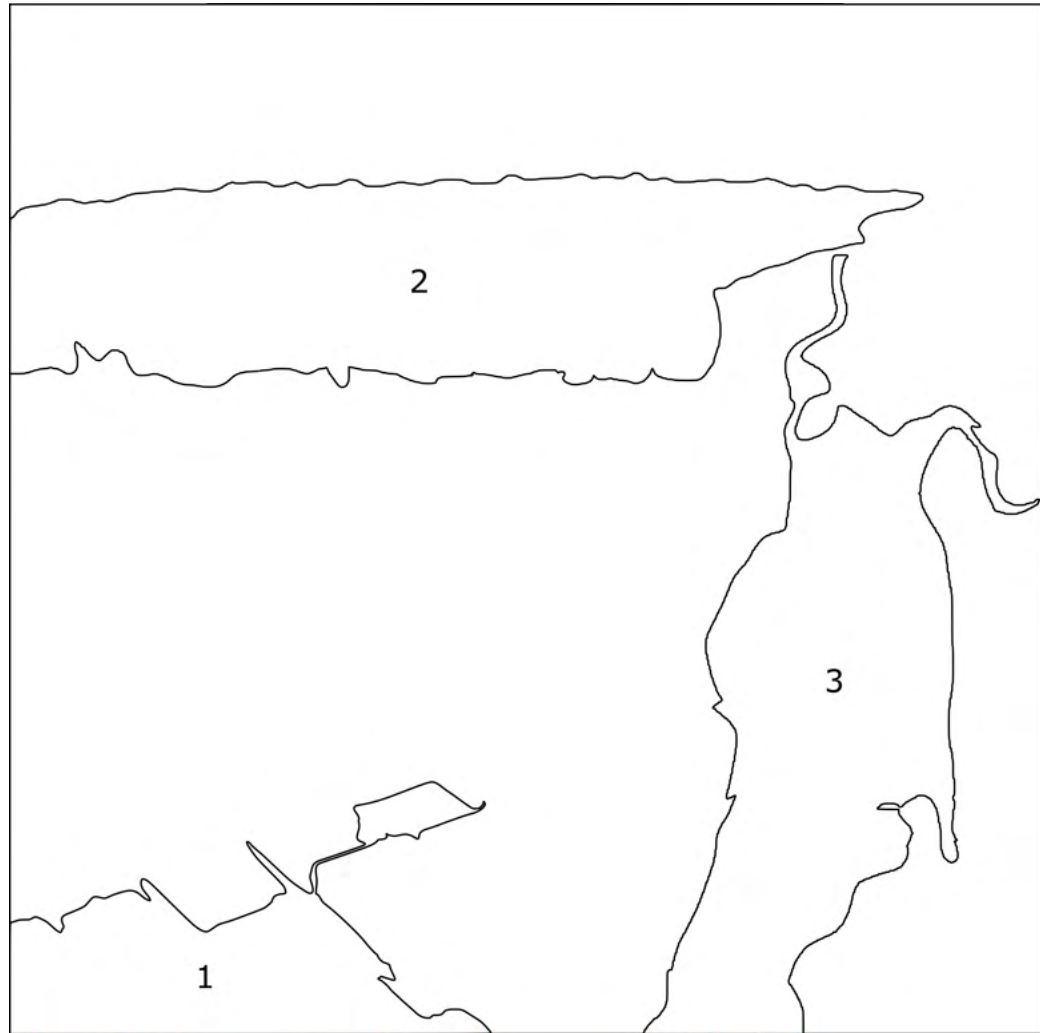


Figure 6: Image generated from fragments of backlink imagery via inpainting



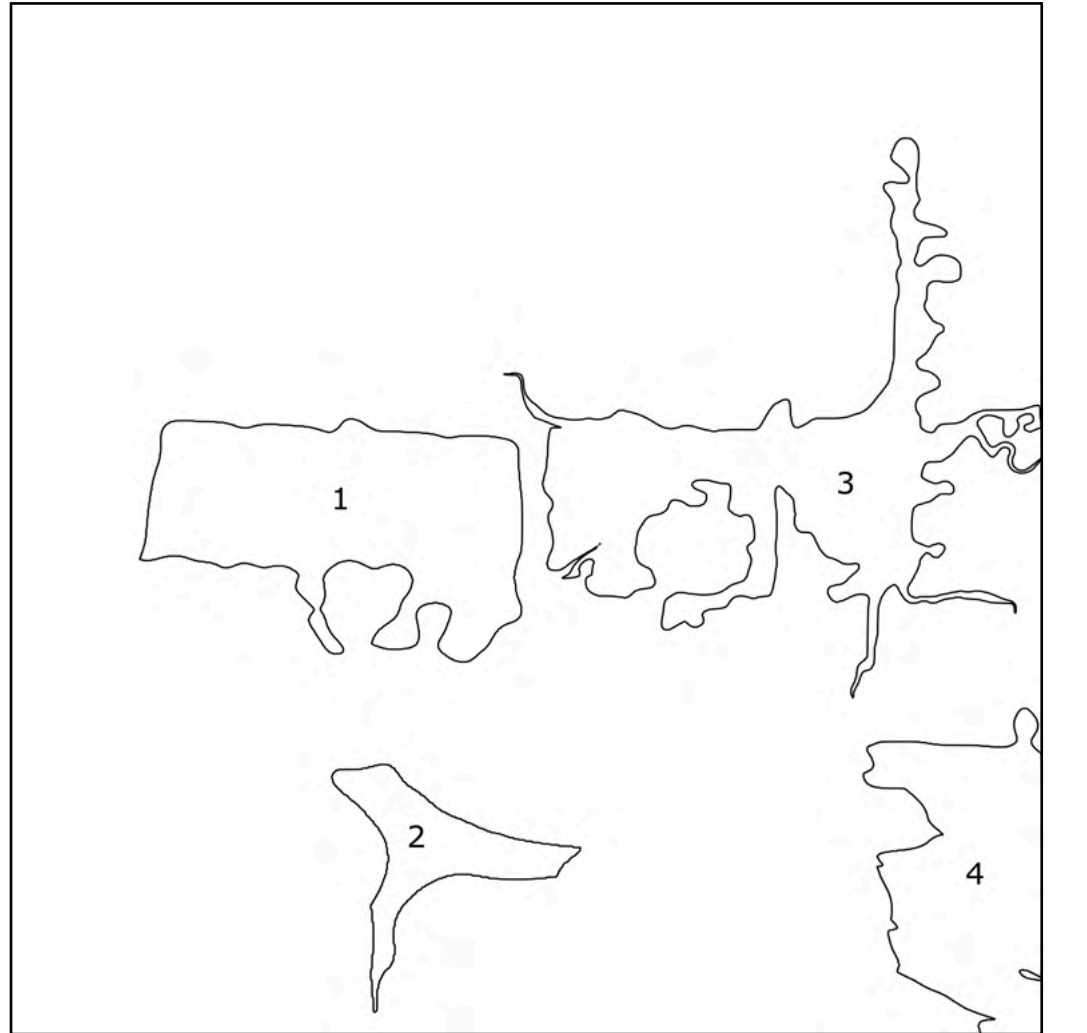
'Extremophile' backlinks:
 1, 2, 6 - 'Hot Spring'
 4 - 'International Space Station'
 3, 5 - 'Spirochaeta americana'



'Deception' backlinks:
 1 - 'Scrabble'
 2 - 'Signal Intelligence'
 3 - 'Secrecy'



'Unmanned aerial vehicle' backlinks:
 1 - 'Walt Disney World'
 2 - 'Lidar'
 3 - 'DARPA'



'Tax' backlinks:
 1,4 - 'Perak'
 2 - 'Biodiesel'
 3 - '610s'

Productive City Hair Care (2022)

‘Productive City Hair Care’ is a proposal detailing how excess human hair within urban areas can be repurposed to improve the viability of food systems and other essential industries.

By 2030, the 5.2 billion humans inhabiting the earth's cities will produce an estimated 400,000 tonnes of hair each year, which equates to 34% of current global clean wool produced by a population of 1 billion sheep yearly. Human's continued treatment of their hair as a waste product calls into question the efficiency of their expanding cities.

Hair is a rich source of keratin and can be used for many other things. Food production, oil spill remediation, medicines, and construction are industries that could be made more vibrant and viable by ingesting hair. The evolution of hair from a material for bodily protection to individual expression makes it all the harder for humans to recognise their hair's potential.

Productive City Hair Care repurposes plugged drains, barber's dustbins and saturated brushes of cities as an efficient patchwork of allotment spaces that comprise an urban farm in anagen. It saves the hair from becoming waste, takes advantage of existing modes of clean-up, and doesn't require expensive equipment or fancy ingredients.

Productive City Hair Care was exhibited at the Tallinn Architecture Biennale 2022 as part of the Future Food Deal Competition through text and illustration (Figure 7). Within this project my contributions involved research, writing, calculations and systems design.

[Link](#)

Collaboration with Johan Michalove, Sasha Smorodina and Hanna Paniutsich



Figure 7: Illustration by Sasha Smorodina

Twilight Fabric (2020-2021)

During the early months of lockdown I became increasingly interested in watching the slow footage from webcams based in random locations around the world. Through a vast network of cameras, every genre of interior space and landscape is represented. This was the gentlest most uneventful form of TV at that moment in time. I wanted to explore ways of capturing the ambience of CCTV footage through representing the subtle shifts in light and colour occurring within these broadcasts.

I developed a number of images that capture the colours within CCTV footage at twilight across different evenings. I became particularly attached to footage from a weather cam at the Calar Alto Observatory in the mountainous desert of Northern Spain (Figure 8). Through experimenting with different colour sorting algorithms and mirroring recordings to create patterns (Figure 9 & 10), the unique oscillations in colour and differences across evenings become more legible. Recordings were captured late into the evening on my Raspberry Pi computer and I enjoyed waking up each morning to see what unique patterns had been generated.

[Link](#)



Figure 8: Calo Alto Observatory web cam

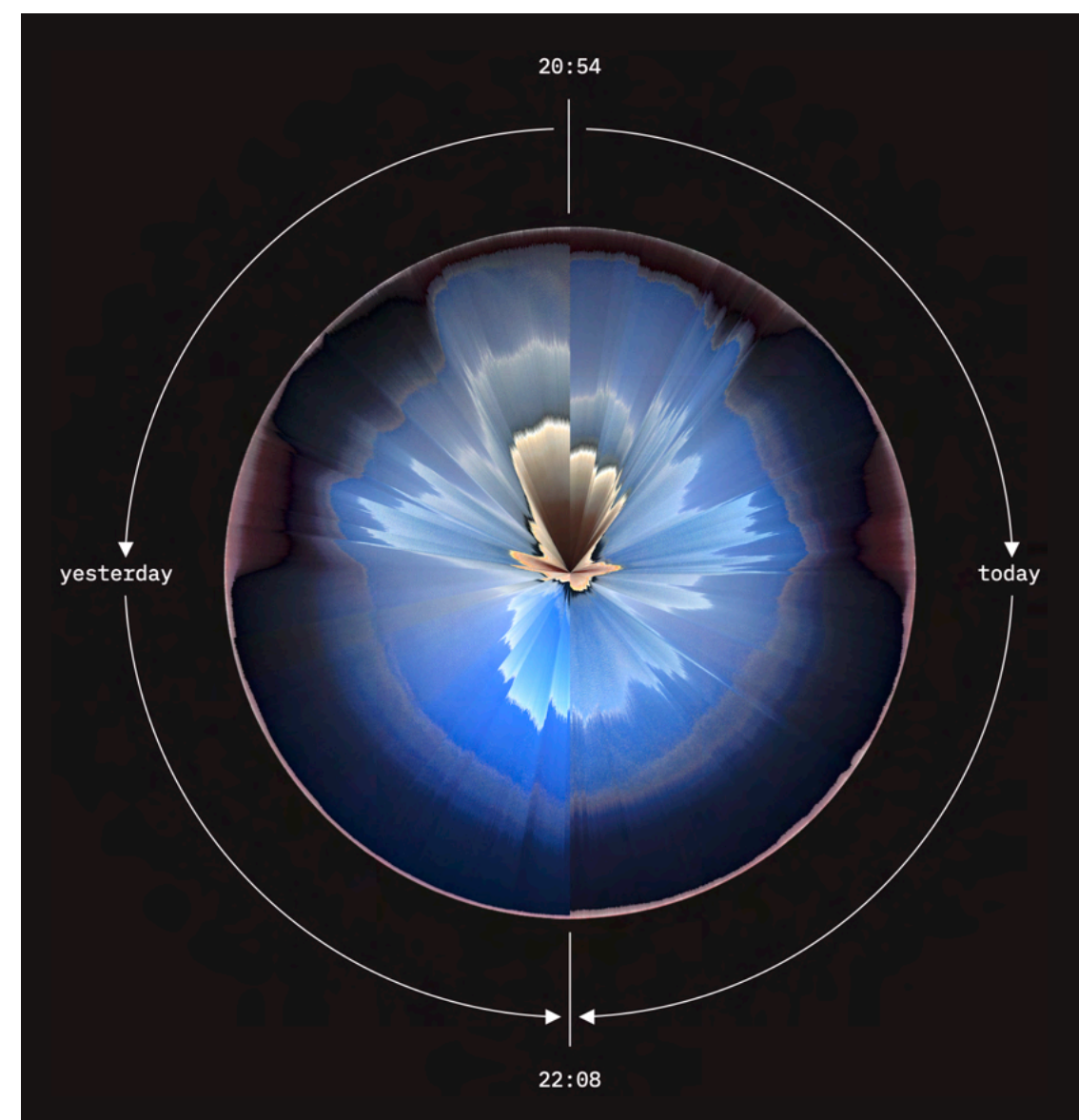


Figure 9: Comparing colour distributions of two evenings

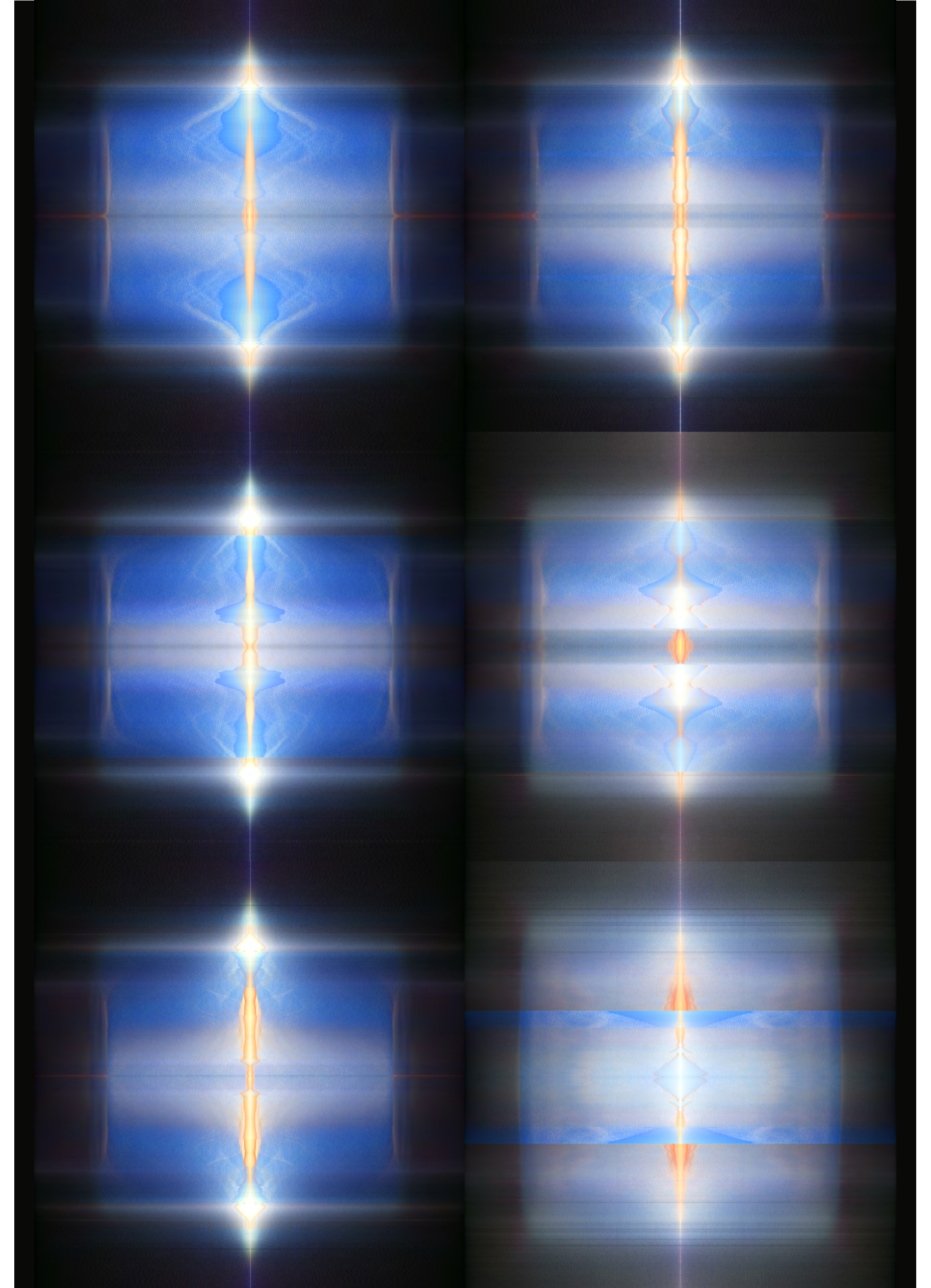


Figure 10: Pattern generated from six evenings

Facial Feedback (2021)

The following portraits are generated from dozens of images of Bella Hadid, Rihanna and Kim Kardashian's doppelgängers. I used dlib (C++ machine learning library) to detect each celebrity's facial landmarks and used these constellations of coordinates as a focal lens to fine-tune and average the faces of their doppelgängers into a collective identity (Figures 11, 12 & 13). This small project arose from my interests in the aesthetic intersections of doppelgängers within science fiction and religion, celebrity culture and social media induced feedback cycles. I am interested in further exploring how platforms mediate monocultures and outliers.



Figure 11: Bella wears Jennifer Lawrence, Carla Bruni, Isabelle Mathers and Joann van den Herik



Figure 12: Rihanna wears Priscila Beatrice, Andele Lara, Renee Kujur, Yna Sertalf and @notavrey



Figure 13: Kim wears Kami Osman, Sonia Ali, Amra Olevic Reyes, Claire Leeson, Naya Rivera, Thalia Almodovar, Jelena Karleuša, Jelena Peric, Larsa Pippen and Monyy Monn

Detecting Tear Gas Canisters with Limited Training Data (2021)

Human rights investigations often require triaging large volumes of open source data in order to find moments within image, or video that are relevant to a given investigation and warrant further inspection. 37-40mm tear gas canisters are some of the most common munitions used against protesters worldwide, however searching for images containing tear gas usage online manually is laborious and time-consuming.

Working with Forensic Architecture and several open-source researchers, we focussed on applying object detection models to facilitate discovery and identification of tear gas canisters for human rights monitors. For convolutional neural network (CNN) based object detection to work, a large amount of training data is required. To achieve our objective, we benchmark methods for training object detectors using limited labelled data: we fine-tune different object detection models on the limited labelled data and compare performance to a few shot detector and augmentation strategies using synthetic data. We provide a dataset for evaluating and training tear gas canister detectors and show how such detectors can be deployed for a real world application such as investigating human rights violations (Figure 14 & 15). Our experiments show that fine-tuning state of the art detectors perform as well as the few shot detector, and including synthetic data can improve results. This research is summarised within a paper that was accepted to the Winter Conference for Applications of Computer Vision 2022.

Within this research my contributions involved training several state of the art detectors (Faster R-CNN and RetinaNet) for benchmarking, data labelling and pre-processing as well as report writing.

Paper

Collaboration with Lachlan Kermode, Chris Tegho and Ashwin D'Cruz



Figure 14: Dataset of tear gas canisters curated by Forensic Architecture containing both real (left) and synthetic images (right)



Figure 15: Detections on test dataset with RetinaNet

Shared Futures (2020)

r/Futurology is a popular subreddit for sharing articles on emerging technologies. I was curious to explore what years are discussed most within articles shared in the subreddit since its inception in 2012.

The Guardian is one of the most popular sources for articles shared in the subreddit. I scraped thousands of articles shared from *the Guardian* and parsed dates from the articles using the Stanford NLP library. The adjacent heat map references the years mentioned within each article against the date the article was shared (Figure 16).

There are clear gravitational pulls to the decade milestones of 2030, 2040 and 2050. These years become important ledges for targets, warnings and roadmaps.

I would like to extend this analysis to a much larger corpus of news on longer timescales to understand political influence and further trends impacting the discussion of future years. I am still considering what potential calendar-based interfaces, social media and content creation could emerge from this approach to cataloguing information.

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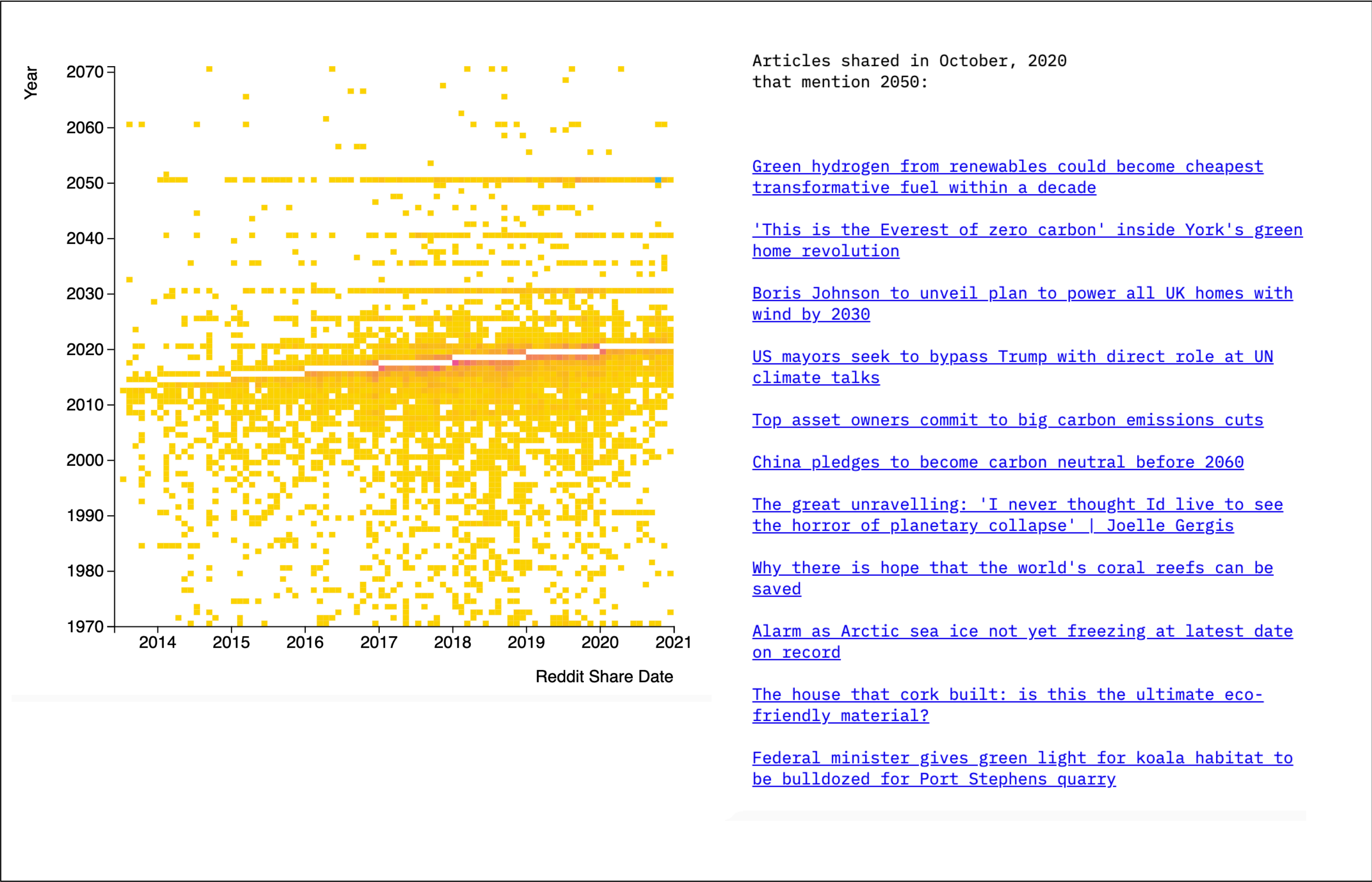


Figure 16: Web interface for navigating temporal map



Figure 17: Review of existing synthetic biology roadmaps from UK government and industry



Bioanarchy in the UK (2017)

In 2017 I collaborated with two other designers researching the current state of synthetic biology in the UK and exploring the potential impact of post-Brexit deregulation and increased uptake of DIY labs. We developed three speculative case studies that detail how different actors impacted by industrial pollution, gentrification and intellectual property theft might utilise synthetic biology as a form of protest and resistance (Figure 17). The project was presented at MOMA NYC as part of the 2017 Biodesign Challenge symposium (Figure 18) and exhibited at the School of Visual Arts.

[Project Article](#)

Collaboration with Joe Revans and Eva Auer

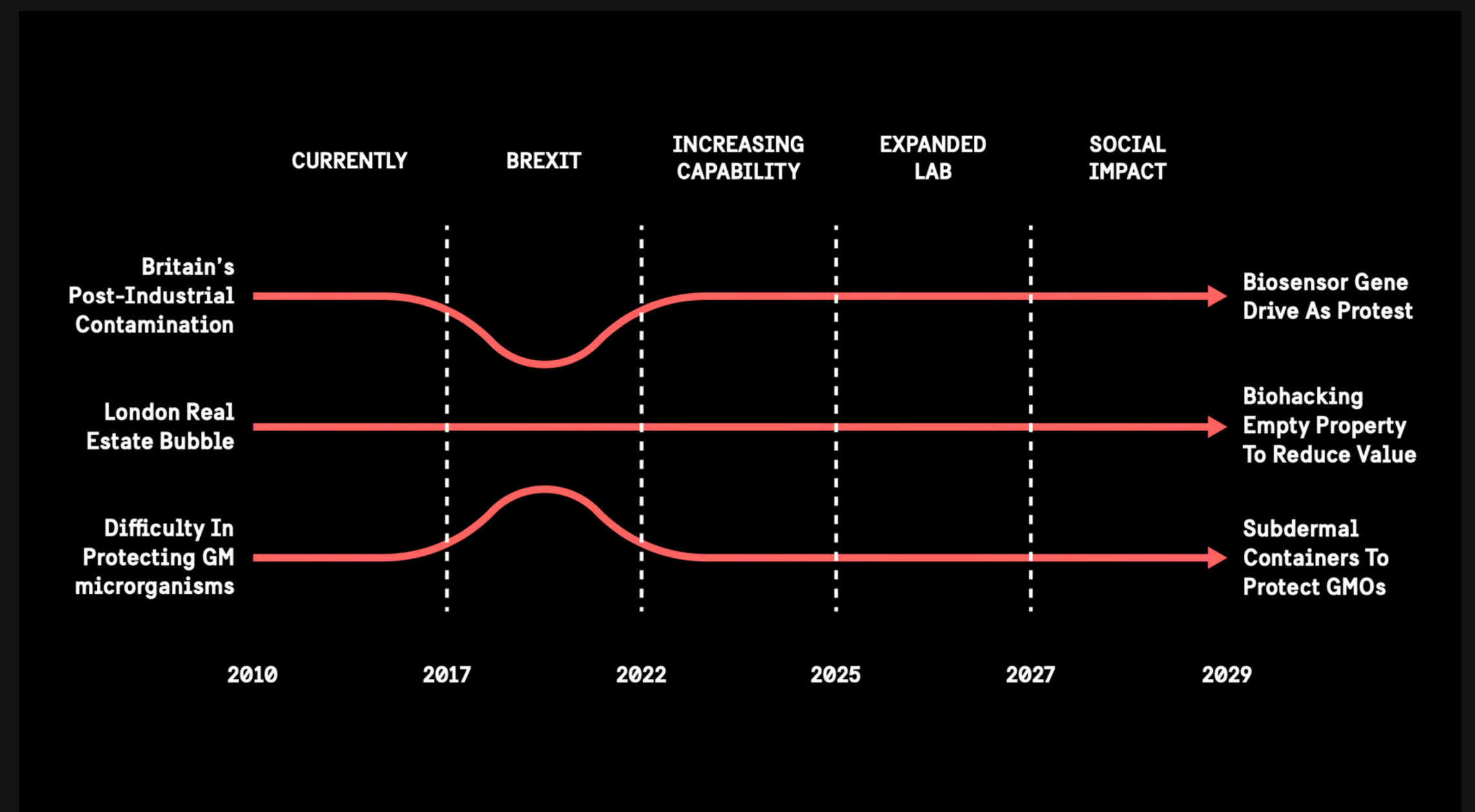


Figure 18: Three speculative case-studies

Biodesign for Social Impact (2017-2018)

Biodesign for Social Impact is a workshop exploring the potential of synthetic biology as a design toolkit to realise alternative political and social goals. Participants are asked to construct alternative worlds founded upon subverted myths and realised through different actors exploiting the affordances of synthetic biology.

The workshop structure is adapted from causal layered analysis, a futures research method that aims to broaden understanding of today’s issues beyond media representation towards underlying myths and metaphors:

- + choose a story of interest from current events (Figure 19)
- + deconstruct the chosen story into actors and then map the interdependence and hierarchies between them
- + analyse the worldview of each actor
- + analyse the origin myths, traumas and cultural beliefs that underpin each worldview

Once the selected real-world phenomena has been dissected to its mythic base, participants are asked to subvert the underlying myth in preparation for world building (Figure 20). Participants are simultaneously introduced to the capabilities of synthetic biology (Figure 21) and asked to consider how select actors might apply these technologies to their own ends. The workshop concludes with participants sharing the different worlds that emerge from their webs of actors in which underlying beliefs have been subverted and novel technologies exploited (Figure 22).

The workshop was held at Dutch Design Week 2017, TU Delft (Industrial Design Engineering) and University of Edinburgh (Design Informatics & Product Design).

Collaboration with Joe Revans and Eva Auer

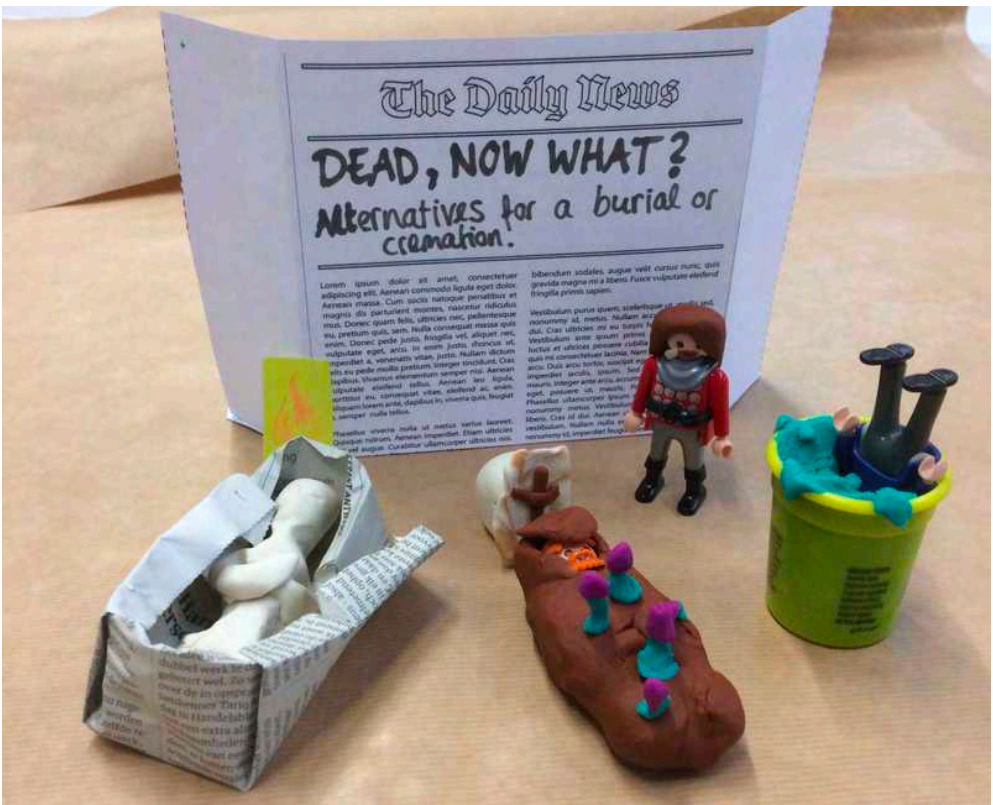


Figure 19: Deconstruction and rebuilding

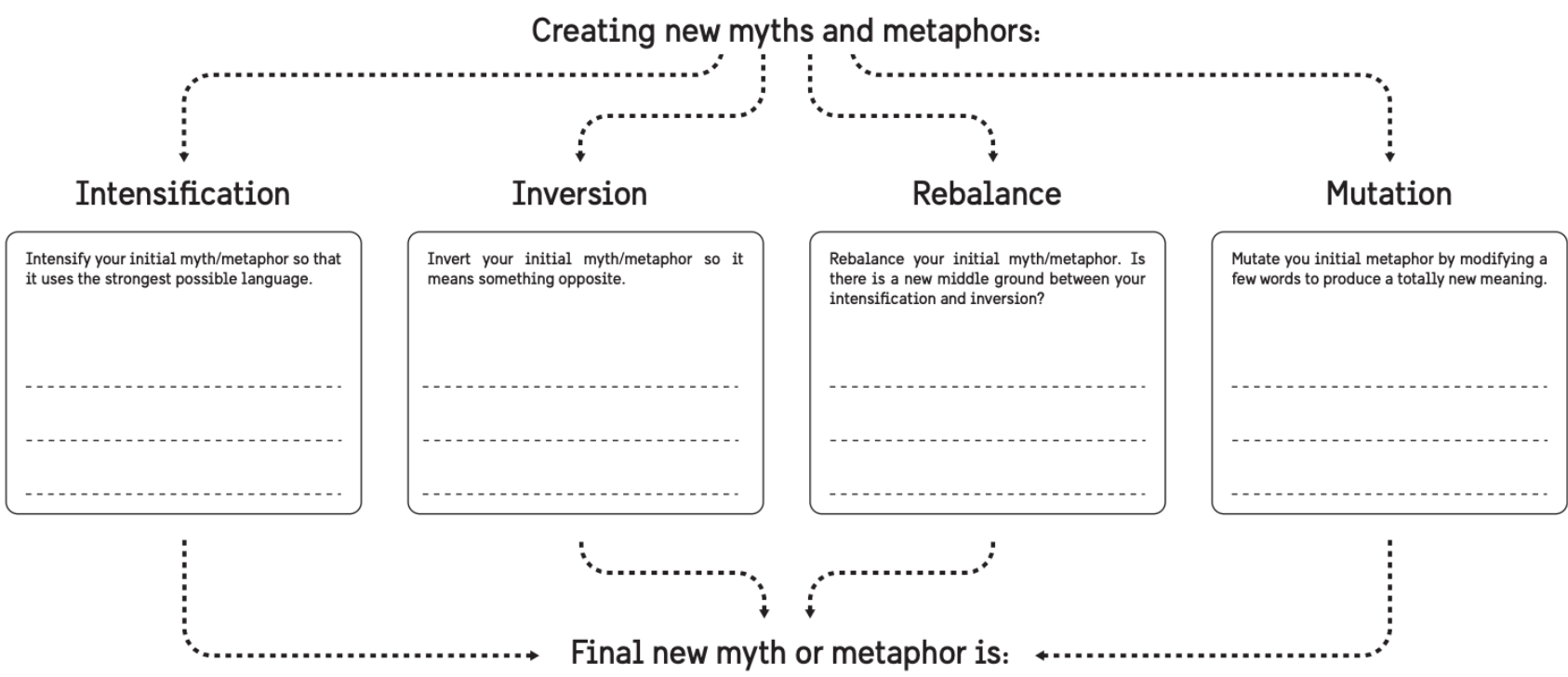


Figure 20: Myth subversion

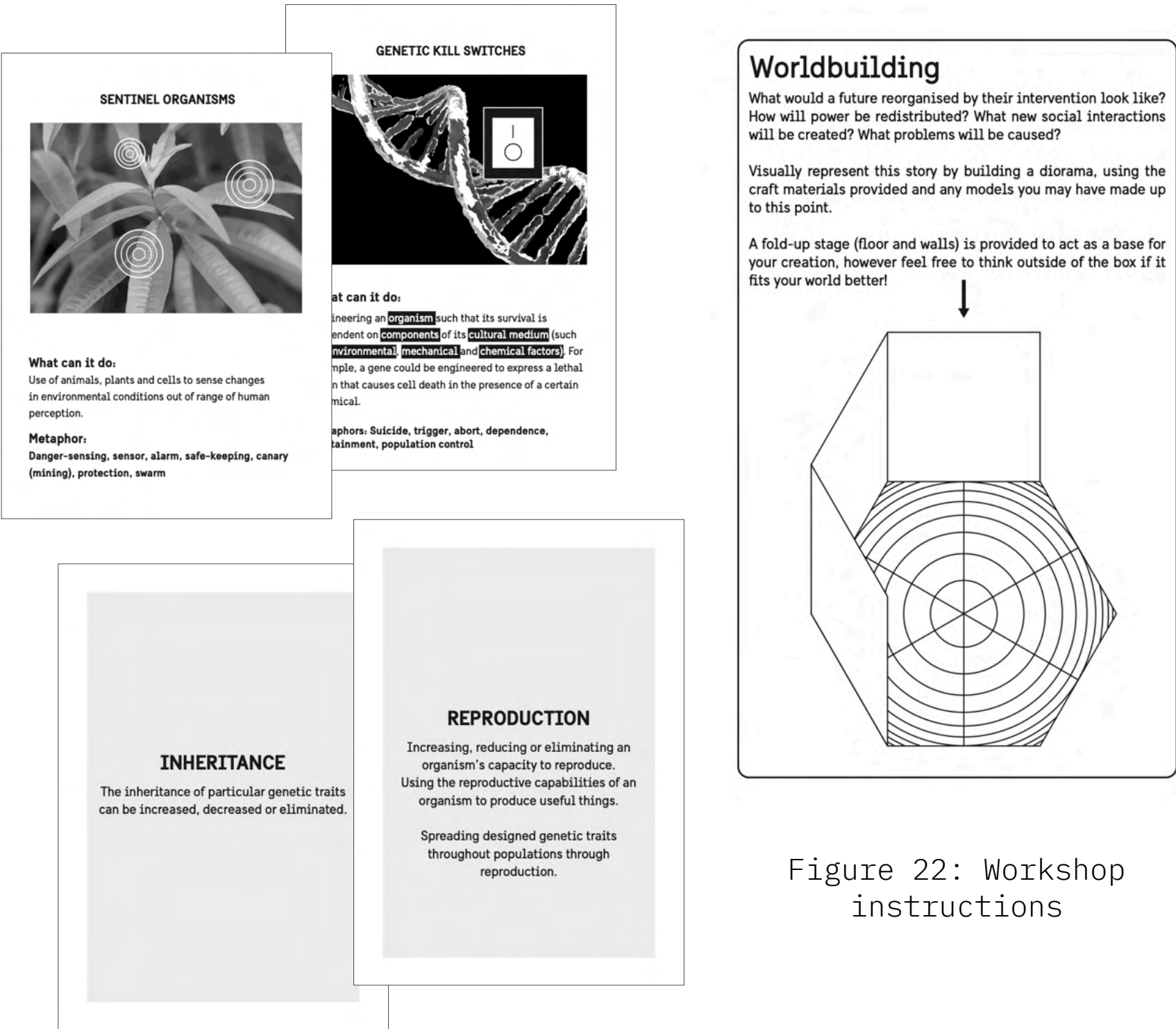


Figure 22: Workshop instructions

Figure 21: Synthetic biology’s technologies and design affordances

Fossil Forest (2021)

Single use plastic water bottles are the ghostly artefacts of vast supply chains constructed upon myths of purity. Despite their limited utility, they are grimly charismatic objects and can be found everywhere. There exists myriad designs and each bottle has playful sonic properties.

Inspired by Jugaad robotics, I designed a tree-like structure that reacts to the environmental conditions where bottled water is sourced (Figure 25). Built from bottles, computer fans, servos and sewage piping, each branch represents a location where water is bottled and generates sound responding to rainfall conditions using the OpenWeatherMap API.

Dry conditions are sonified through the creaking of steel wire across the knurls of a bottle cap echoing through the chamber of an empty bottle (Figure 24). These servo driven noises are an attempt to mimic dry trees creaking. Rainy conditions are sonified through the dull moan of airflow over a slit cut into the side of an empty bottle (Figure 22). With enough time I would build an entire forest of these trees to orchestrate the sounds of bottles from locations around the world.

[Link](#)



Figure 22: Wind through a crack in the window during a rainstorm



Figure 24: Creaking of dry wood in a quiet forest



Figure 25: Tree with branches monitoring rainfall at Glorat Estate, East Dumbartonshire (top-left), Buxton, Derbyshire (right) and Broughton, Hampshire (bottom-left)

Ambient Twitcher (2020-2021)

Inspired by CAPTCHA systems used to train machine learning algorithms to recognise Google street-view objects, I built a simple Chrome extension to train me into recognising the sounds of different species of bird recently spotted near the streets and walkways wherever I happen to be in the world (Figure 28). The extension (Figure 26) randomly appears as a pop-up throughout the day and encourages me to guess the species of bird through listening to it's song and exploring its location. Panoramas are downloaded via the Google Maps Javascript API allowing me to wander through each location before submitting my answer. This extension is helping me to build a general awareness of the UK's green spaces and nature whilst I'm logged on. I'm continuing to explore what other kinds of exploration I could build via maps, physical computing devices (Figure 27), sound libraries and other objects tagged with geospatial data.

[Link](#)

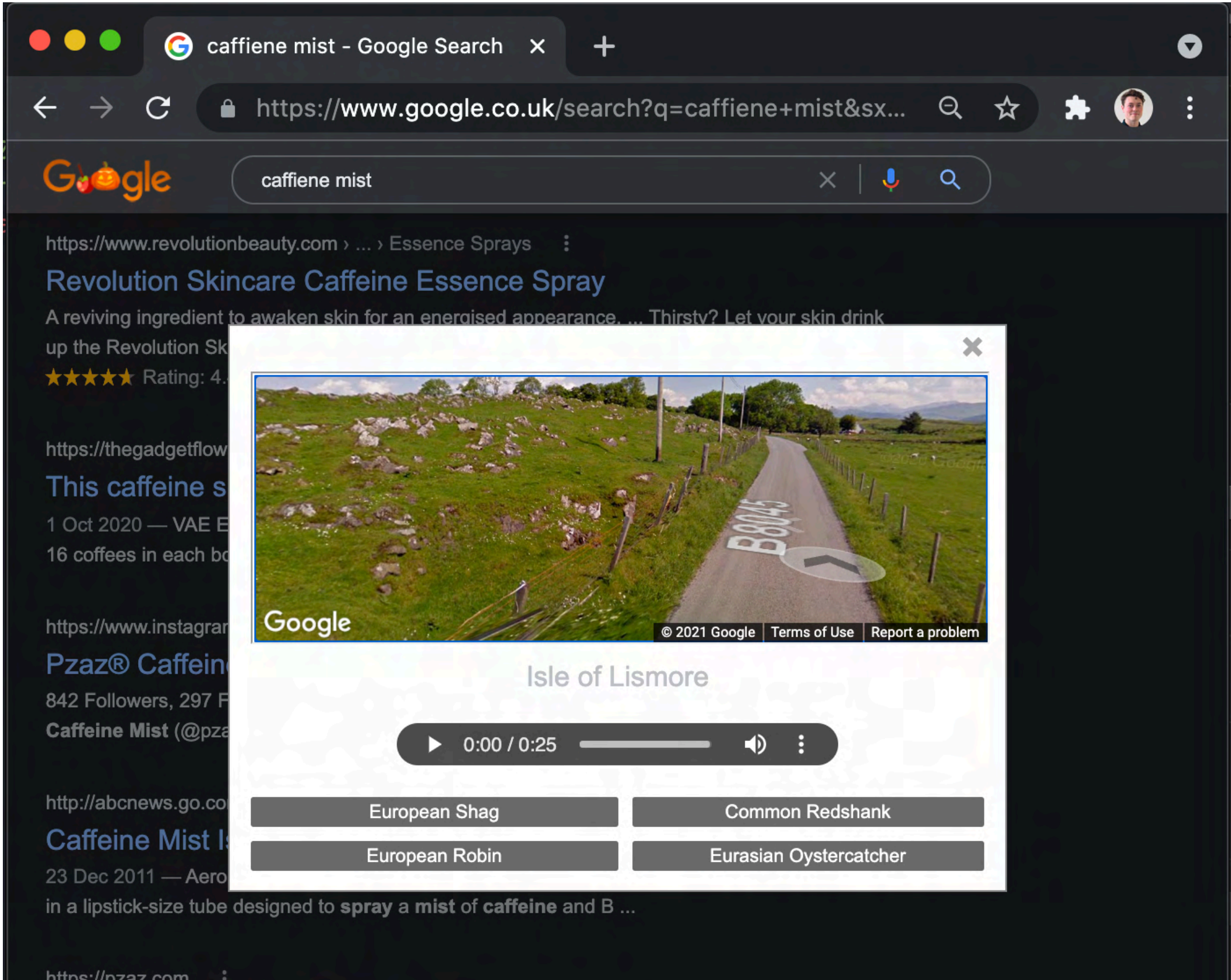


Figure 26: Ambient Twitcher chrome extension

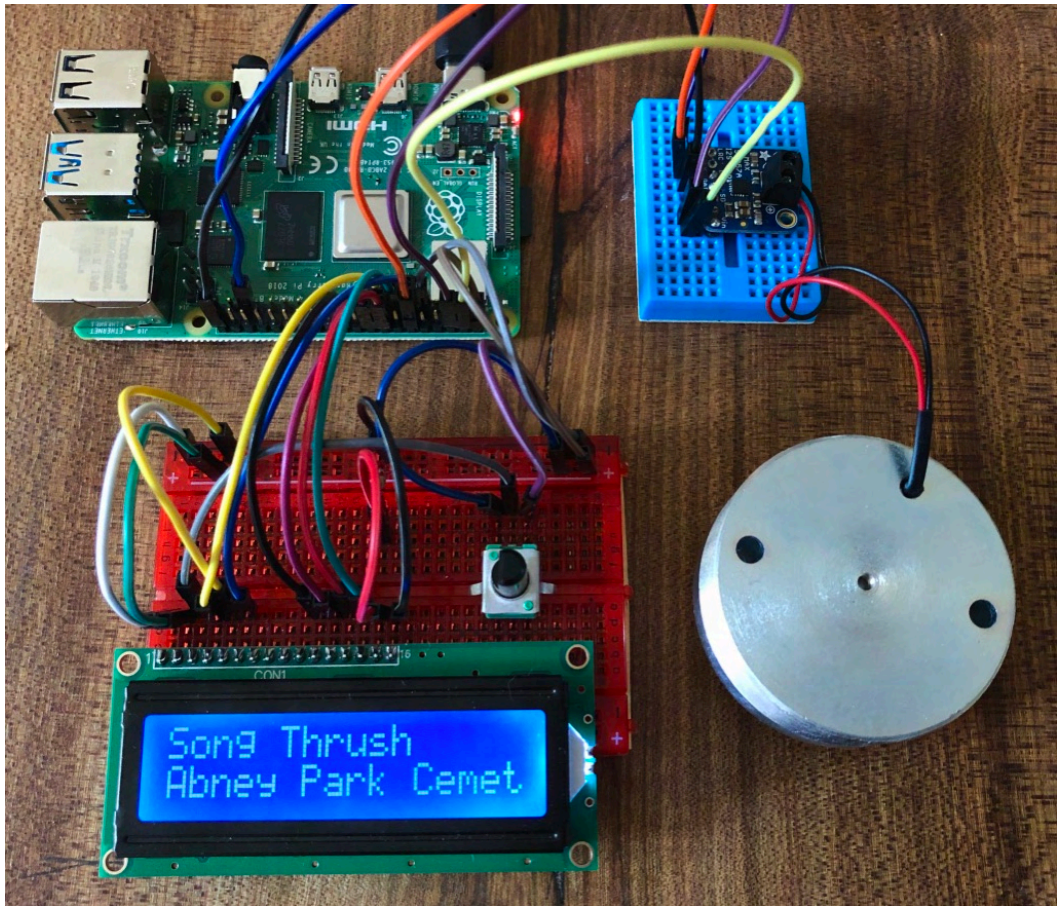


Figure 27: Bird song audio device



Figure 28: Bird sighting logged to eBird

Solutions Engineering at Seldon (2021-2022)

Seldon is an MLOps (machine learning operations) start-up building open-source software to deploy, serve and manage machine learning models faster and with reduced risk. Seldon has raised \$10 million to date and is being used by over 60% of Fortune 500 companies.

I worked within a global team of three Solutions Engineers tasked with managing proof-of-concepts, delivering architecture deep dives and building code patterns tied to client challenges. This role required me to navigate Seldon's open-source cloud-native software stack (Figure 29) alongside popular data science tooling (Python, Tensorflow, scikit-learn etc.)

Speaking daily to clients across industries and geographies gave me insight into the hidden technical debt encountered when trying to productionise machine learning at scale. Scalability and deployment speed are often hindered by siloed data science and DevOps teams. Model performance degradation and lack of compliance without monitoring are common challenges encountered within the 'last mile of data science'.

As an example engagement, I worked with Noitso, the Danish financial consulting company, as lead solutions engineer throughout the pre-sales cycle to the point of them becoming a customer. Working closely with their data science team, I was able to demonstrate the value of Seldon's monitoring and explainability within their machine learning operations, tailored to their specific population datasets and credit optimisation use-cases.

[Code Case Study](#)

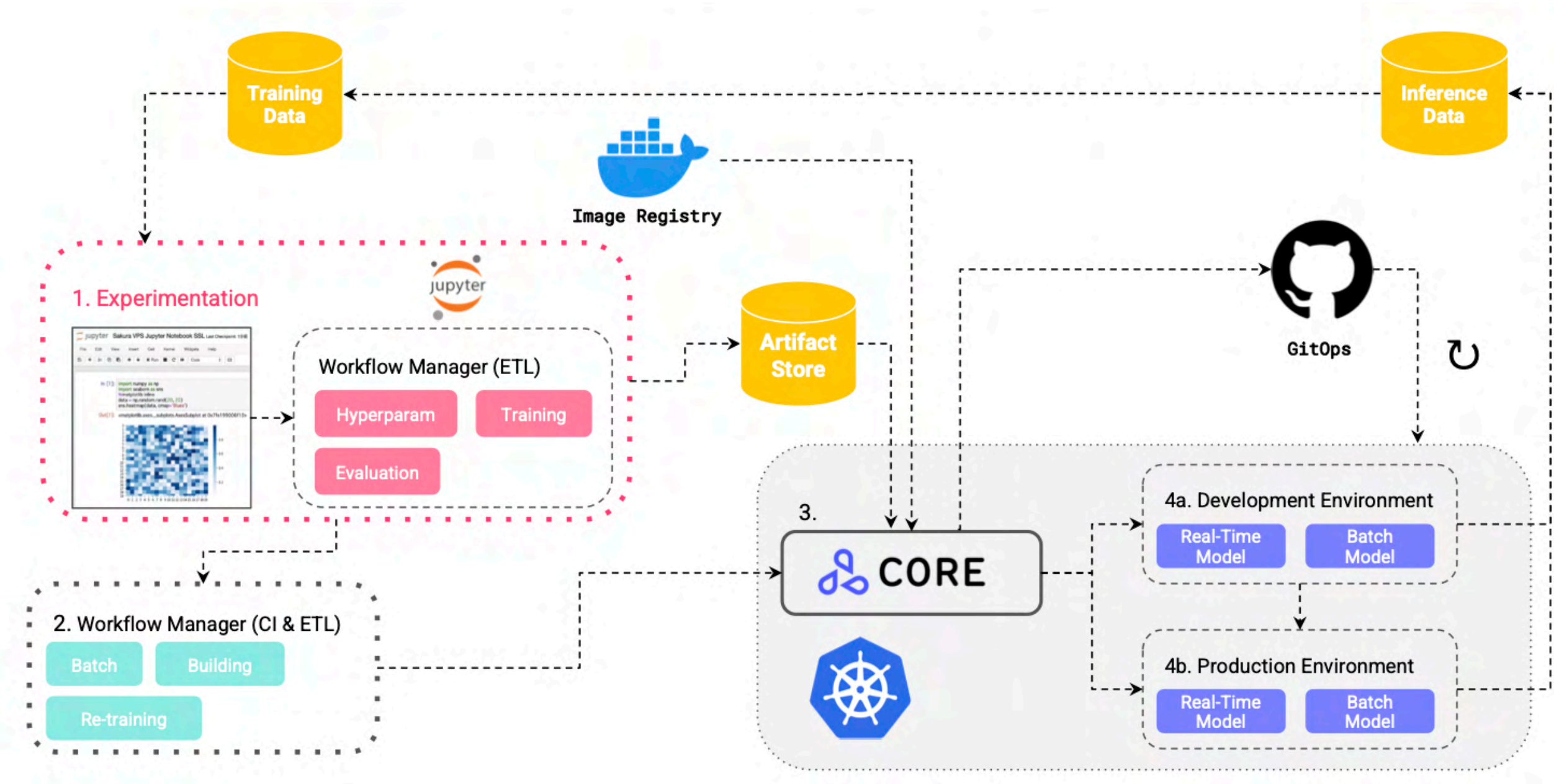


Figure 29: Machine learning deployment lifecycle

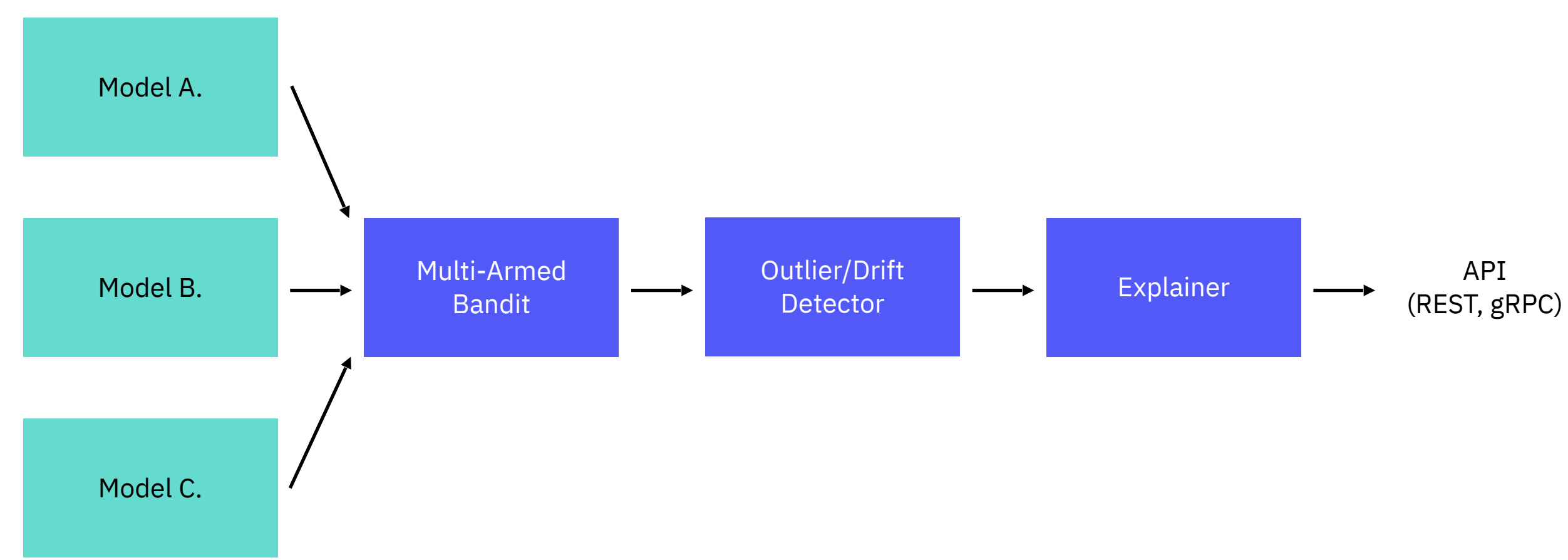


Figure 20: Complex Inference Graph